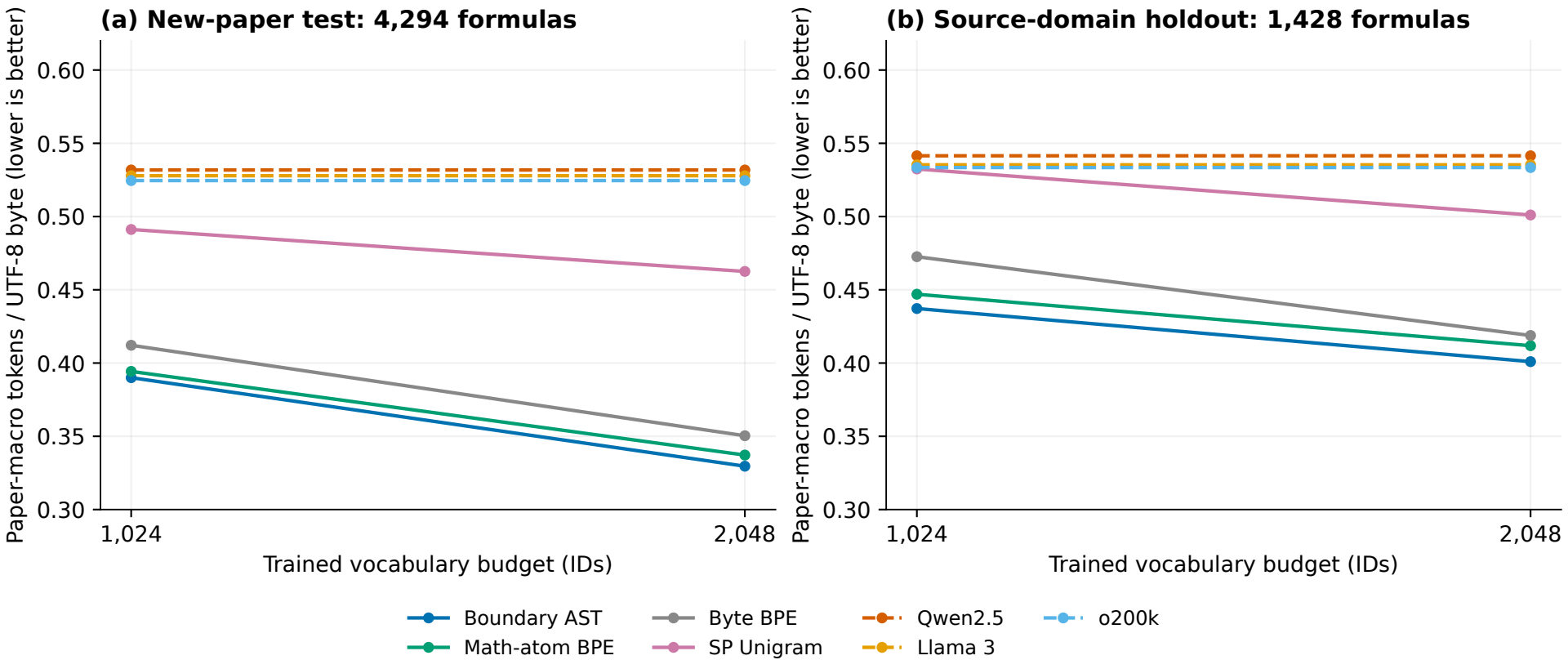
(a) New-paper test: 4,294 formulas
(b) Source-domain holdout: 1,428 formulas
Paper-macro tokens / UTF-8 byte (lower is better)
0.60
0.55
0.50
0.45
0.40
0.35
0.30
1,024
2,048
Trained vocabulary budget (IDs)
Boundary AST
Math-atom BPE
Byte BPE
SP Unigram
Qwen2.5
Llama 3
o200k